CYBORG SUPERMARKET ENTERPRISE DOCUMENTATION

Database ERD and Data Dictionary

Complete Schema Specifications, Constraints, and Index Strategy

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Executive Publication Record

Organization:	Cyborg Omnichannel Engineering Organization
Ecosystem:	Windows POS • Android Mobile • Web Storefront • Admin Web • Backend
Primary Authority:	cyborg_system_architecture.pdf (16-Page Blueprint)
Source Dataset:	super_market(1).xlsx (1,000 Verified Egyptian Retail Records)
Security Level:	Enterprise Retail Distribution / Stakeholder Review
System Stack:	NestJS • TypeScript • PostgreSQL 16 • Redis 7 • SQLite • Flutter
Target Currency:	Egyptian Pounds (EGP) • Cairo Timezone (UTC+2 / UTC+3)

1. Entity-Relationship Overview

The relational database schema is normalized to 3NF while providing fast index paths for high-frequency retail checkout and catalog search:

Table Name	Primary Key	Foreign Keys	Description & Constraints
users	id (UUID)	None	Identity, password hash, hiera
categories	id (INT)	parent_id -> categories.id	Self-referencing taxonomy tree
products	id (INT)	category_id, subcategory_id	1,000 real SKUs, indexed barco
inventory	id (INT)	product_id -> products.id	Stock quantity, reservations,
shifts	id (UUID)	cashier_id -> users.id	Cash drawer reconciliation and
orders	id (UUID)	user_id, shift_id	Financial order ledger with id
order_items	id (INT)	order_id, product_id	Snapshot product prices and qu
carts	id (UUID)	user_id	Personal & family carts with s
sync_events	id (INT)	None	Cloud-local sync event log wit

2. Data Dictionary ? Products Table

Column	Data Type	Nullable	Description / Source Mapping
id	INTEGER PK	NO	Unique sequential product ID (
barcode	VARCHAR(32)	NO	Indexed product barcode from E
name_ar	VARCHAR(255)	NO	Arabic product name (’ArabicPr
name_en	VARCHAR(255)	NO	English product name (’English
category_id	INTEGER FK	NO	References categories.id (Fres
brand	VARCHAR(100)	YES	Product brand name (143 distin
price	NUMERIC(10,2)	NO	Exact price in Egyptian Pounds
is_active	INTEGER	NO	Boolean flag (1=Active, 0=Deac

3. Indexing & Query Optimization Strategy

- idx_products_barcode: Fast B-Tree index for sub-5ms barcode scans on cashier terminals.

- idx_inventory_product_store: Composite index on (product_id, store_id) for rapid concurrency checks.

- idx_orders_status_date: Indexes for dashboard analytics and cashier shift summaries.